



Combat Robotics Out West (CROW)



**Visual Rulebook V1
2026**

Modified version of SPARC Robot Construction Specifications v1.3,
SPARC Match Rules v1.3, and NHRL rules.

Overview

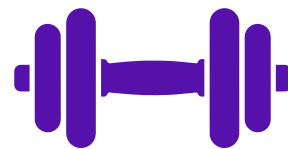
Outline

This document outlines the ruleset for constructing and fighting 150g antweight combat robots at Combat Robotics Out West. It encourages creative designs, exciting fights, and a level playing field while keeping everyone safe. [Full ruleset can be found on GitHub.](#)



Fight Basics

- 3 minute matches
- Win by KO, Rifting, or judges decision



Weight

150 grams



Safety Basics

- Must have a **switch** to turn off power without disassembling the robot and a visible power **LED**.
- Your **battery** must be **covered**.
- Must pass a **radio failsafe** test.
- Must have a **weapon lock** that prevents any spinning weapons from moving while outside the arena.

League

Combat Robotics Out West has two leagues. Each robot may only compete in one league. See 'Tournament' for limits on entry qty.



Plastic Ant League

Class Intent

Designed as an accessible entry-point to the sport, the destructive plastic ant league has limitations on the types of materials and manufacturing methods you can use for your 150g robot.

This class levels the playing field for new builders while forcing veterans to solve problems in new ways within the restrictions.

Beginners are encouraged to compete by lowered costs and moderate destructiveness.

Open Ant League

The destructive **open antweight** league has no restrictions on build materials.

Expect metal weapons and maximum destruction from these 150g robots.

Safety

To compete, robots must pass safety. To pass safety your robot must include the following:



Switch

You must be able to turn off power without disassembling the robot. Switches cannot shift or become inaccessible during a fight. Rigidly mount them to your chassis!



LED

An LED must be easily visible from the side of the arena when your robot is powered on. Your receiver / control board LED is NOT enough.



Radio Failsafe

Upon loss of radio signal (e.g. turning off your transmitter), all bot motion / weapons stop automatically. Bluetooth robot control for a bot with a spinning weapon is NOT permitted.



Weapon Lock

Any spinning weapon must have a physical lock to prevent it from activating accidentally. These must not rely on delicate robot geometry. E.g. a squeeze clamp or bolt is suitable



Event

Weapons must not spin up outside a test-box or the arena. **EVER.** if testing drive, weapon must be physically locked and wheels propped off the ground.



Covered Battery

You must not be able to see the battery and it must be reasonably protected.

Build Restrictions

We restrict weapons for three reasons. To keep fights entertaining, make sure everyone outside the arena stays safe, and to ensure our arena doesn't get too wrecked.



Projectiles



Entanglement



Fire



Liquid



EMPs



Radio Jamming



High Voltage



Light or Smoke



Explosives

If you have a weapon that may exploit a loophole or if you're unsure, contact the event organiser to verify if it's allowed. We don't want to have to disqualify anyone.

Weight

There may be weight bonuses available for non-standard forms of movement. To qualify for a weight bonus, your weapon must not be part of the movement system.

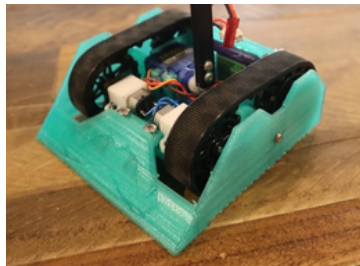
Standard (150g)

The vast majority of antweight robots are **150 grams**. Your bot will be weighted during safety. If it is overweight you will not be allowed to compete. No leeway will be provided under any circumstances.

**Examples of standard motion systems
that *DO NOT* get a weight bonus:**



Wheels



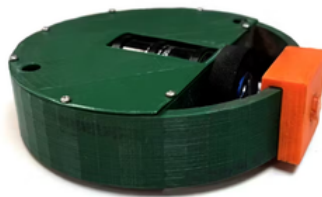
Tank tracks



Gyro walker



Bristle bot



Melty brain



Torque walker

Weight

Weight bonuses should be utilised for unique locomotion. Robots must show that at least half of the weight bonus applied for is used in moving parts of the locomotion mechanism, not including motors or off-the-shelf gearboxes. Flying/Hovering robots exempt.

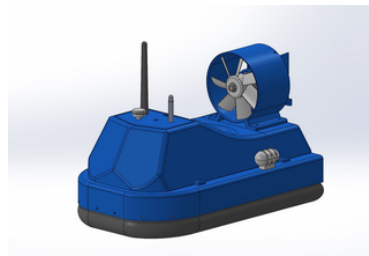
Shufflers (225g)

Shufflers have no rolling elements in contact with the ground. This qualifies them for up to a 50% weight bonus, which means they can be up to 225g.

Examples of robots that qualify for the shuffler weight bonus:



Cam shuffler



Hovercrafts

Walkers (300g)

Bots that can fly or walk qualify for a 100% weight bonus which means they can be up to 300g. “Walk” is defined as having two or more non-continuous actuated joints within each leg.



Drone



Walker



Plastic Ant Rules

These are the adjustments made for the plastic class on the open class ruleset. Class intent and safety is priority.

Allowed materials:

PLA, PLA+, ABS, PET, PET-G, Acrylic, Cardboard, Plywood and MDF are permitted and should be the primary materials in your robot. Unlisted materials may be permitted if they follow class intent, check with the organisers ahead of time. Electronics are exempt.

- Decorations of any material are permitted so long as they are not used as armor, impactors, or structural features.
- Unmodified metal fasteners **can** be used in mechanisms as axles or for fixing parts together.
 - A *Fastener* is defined as:
 - An off-the-shelf part designed to affix two or more components together by means of a thread
 - Must be easily obtainable from a common hardware store or online vendor
 - Examples include jaycar, bunnings, Aliexpress, local hardware store
 - Unit value of 5AUD or less
 - Washers are OK
 - If they pass through multiple points in a chassis, it is understood that this inherently adds some strength and is permitted so long as it is not abused.
 - Bearings and bushings are similarly permitted.



Plastic Ant Rules

Allowed materials cont.:

- Grippy materials can be used as wheel tread or tracks. This includes rubber, silicon, resins etc. Rope, springs, elastics can be used in actuators or for power transmission only.
- Heat set inserts **are** permitted
- Foam is only permitted as a weapon material.
- Cheap recycled goods such as takeout containers are permitted.

Components:

Any off-the-shelf or custom electronics are permitted. This includes sensors, motor controllers, microcontrollers or PCBs. Use best intent to prevent short-circuits and fires.

Robots **can** use any off-the-shelf motors and gearboxes. Components **can** be disassembled, swapped or interchanged as desired. These **cannot** be modified by cutting, drilling or other destructive process, wiring exempt.



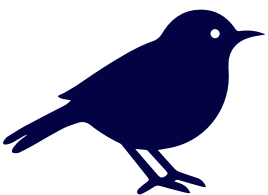
Tournament

Robots will first be required to pass safety checks before they can compete. Do your best to ensure your robot is safe *before* arriving at the event!

Robots will be registered on the day under their driver's name. Only two entry can be signed up per *driver*, per *league*. Examples of this:

- ✓ Parent builds and enters three plastic ant robots; one for them to drive, one for each of their children
- ✓ A builder enters and drives two plastic ants and one open ant
- ✓ A driver who already has an entry is recruited to operate one part of another's multibot
- ✗ Three or more open ants are planned to be driven by one builder

CROW Tournaments are split into two parts. These will be independent for both PLANT and Open leagues.



“Round Robin” - randomly play other robots to gain points! You will be seeded in the tournament based on your points. Amount of fights will be decided based on number of participants on the day

Win 1 points
Loss 0 points



Tournament - single elimination bracket including finals, semi finals, quarter finals etc until a winner is established!

Tournament

Repairs

When your robot takes damage during a fight, you will have some time between matches and over breaks to repair it. Facilities and other builders may be available to assist but try to be self-sufficient in case equipment is unavailable.

A builder can bring as many spare parts as they want to repair their machines but *cannot* substitute complete robots. This includes differing configurations. Components or mechanisms should be swapped out individually, understanding that the machine will likely become a 'Ship of Theseus'.

Load in procedure

1 Robot placed
in arena

2 Robot turned
on via switch

3 Weapon lock
removed

Repeat for every robot

4



**DOOR
CLOSED**

5

**Radio
Transmitters
turned on**

Load out procedure is the same but in reverse.

Depending on robot design it may be deemed safer to switch step 2 and 3 during load out and in, this is at staff discretion

Matches

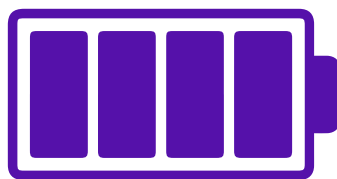
To start a match you will be counted in from 3. Matches last 3 minutes.

Knock Out (KO)

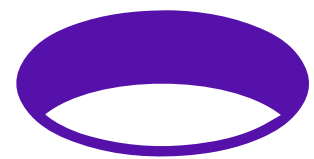
A robot is knocked out and loses if:



**No
motion**



**Battery
Exposed**



**Lifted (gets
in the out of
bounds area)**

Judges decision

A judge decides the winner if both bots survive 3 minutes. The following factors influence the decision:

Damage

Broken drive,
wheels, weapon,
armour etc.

Aggression

Who was initiating
attacks rather than
running away?

Control

Pins, pushing,
flipping, etc.

Tapping out

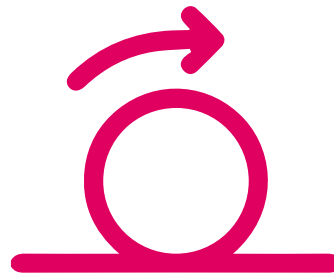
If you want to prevent further damage to your robot, you may announce you're tapping out at any point during a match (usually accompanied by banging on the arena wall). This gives your opponent the win.

Mobility

Robots must have controlled motion. You must be able to drive from the door to a starting corner in under 30 seconds.

Show motion

If the referee calls for you to show motion during a match, you have 10 seconds to move away from your current location, else be counted out.

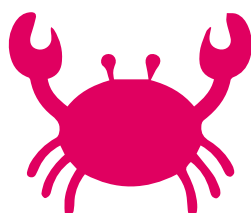


Pins

Pinning another bot means you are preventing it from moving. This shows control. You can only pin another bot for 10 seconds before release.

Unstick

If two bots get stuck together, the match will be paused, controllers turned off, and a referee will separate the two robots so the match can be resumed. After three unsticks the fight goes to a judges decision



Crabwalking

If one side of your drive system gets damaged, crabwalking is a valid form of motion. However, if you're just going in circles you will be counted out.

Combat Robotics Out West



ARTIFACTORY
PERTH'S MAKER SPACE

We would like to extend a massive thanks
to the Robot Royale team for allowing us
to use their rule-sets and designs.